

AI-Based Personalized Fitness And Yoga Recommendation System

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Abstract—Objective:

In recent years, the importance of maintaining fitness and wellness has grown significantly due to the rise in inactive lifestyles, increased stress, obesity, and chronic health issues. People today are actively looking for fitness solutions that are customized according to their personal health needs rather than following the traditional “one-size-fits-all” programs, which often ignore critical differences in body types, health history, activity levels, and individual goals. Each person is unique, and these variations call for a fitness strategy that is more personal and focused.

To address this gap, our research introduces an AI-Based Personalized Fitness and Yoga Recommendation System. This system intelligently creates exercise and yoga schedules based on user-provided details like weight, age, health concerns (such as back pain, diabetes, high blood pressure, stress, insomnia, posture problems, and limited mobility), current fitness level (beginner, intermediate, or advanced), main fitness objectives (e.g., improving fitness, building strength, losing weight, increasing flexibility, or managing stress), and daily activity levels.

The proposed system aims to boost user participation by offering fitness routines that are safe, effective, and specifically designed for each individual's physical and mental health needs. It also integrates yoga practices focused on mental well-being along with physical workouts. This paper details the system's framework, the AI methods used, its web-based deployment, user testing results, and the potential it holds for future digital health technologies. Overall, this work highlights how personalized AI models can significantly change the way fitness and wellness programs are delivered across the world.

General Terms

Fitness, wellness, Personalized Health, Artificial Intelligence, Recommendation System, Yoga, Exercise.

Keywords

AI-based fitness, Personalized exercise plan, Customized yoga routines, Stress management, Machine Learning.

I. INTRODUCTION

In today's digital age, the focus on health and wellness has intensified as more people are experiencing the negative effects of sedentary living, rising stress levels, and unhealthy lifestyle habits[2]. This has led to a noticeable increase in chronic illnesses. Traditional fitness and yoga programs tend to apply a general approach to all participants, overlooking the critical personal differences like age, weight, pre-existing medical conditions, fitness abilities, and personal fitness goals. As a result, these conventional methods often fail to meet individual needs effectively or safely.

Personalized fitness plans are essential to ensure maximum efficiency, better safety, and sustained motivation. By considering unique individual profiles, users are more likely to follow routines consistently and achieve their health targets.

Artificial Intelligence (AI) provides an innovative way to make fitness and yoga plans dynamic and customized. AI can process complex user data and suggest scientifically backed workout and yoga schedules tailored to each person's needs. By applying AI algorithms, we can create systems that not only take personal preferences into account but also recommend the most effective pathways to wellness goals.

This paper discusses the development and execution of an AI-powered system designed to generate customized fitness and yoga plans based on user inputs such as age, weight, health conditions, activity levels, and fitness aspirations.

i. Problem Statement:

Traditional fitness and yoga programs often fail to address individual differences such as age, health conditions, fitness levels, and personal goals. This one-size-fits-all approach can lead to ineffective results or even pose health risks. Many users struggle to find routines suited to their needs, which lowers motivation and consistency in following fitness plans. There is a clear need for a system that can generate safe, personalized fitness and yoga recommendations, ensuring better engagement, safety, and long-term success for every individual.

ii. Objectives:

The main goals of this project are as follows:

1. To design and build an AI-driven system that offers personalized fitness and yoga recommendations.
2. To consider various user inputs including weight, age, existing health issues, activity levels, fitness stages, and specific goals while generating fitness plans.

3. To ensure that the recommended exercises and yoga practices are both effective and safe for users, especially those with particular health concerns.
4. To enhance user engagement, help users achieve their fitness goals more efficiently, and promote the adoption of long-term healthy habits.

iii. Expected Outcomes:

If implemented successfully, the AI-Based Personalized Fitness and Yoga Recommendation System should be able to:

- Generate personalized daily fitness and yoga schedules aligned with each user's unique profile.
- Offer targeted advice and exercise routines for health concerns like back pain, diabetes, hypertension, stress, poor posture, and mobility issues.
- Provide workout plans categorized based on the user's fitness level, whether beginner, intermediate, or advanced.
- Lead to higher satisfaction rates among users and help them achieve their health and fitness goals more effectively.

II. LITERATURE REVIEW

Researchers Several studies and past works underline the significant benefits of personalized fitness programs over traditional, generalized exercise routines.

In a study conducted by Smith et al. [1], it was found that individuals who followed exercise routines customized to their personal needs showed a much higher rate of adherence compared to those following standard workout plans. Participants involved in personalized fitness programs were more motivated, stuck with their routines longer, and experienced better outcomes overall. This research emphasized that tailoring workouts to the individual can dramatically reduce the chances of participants dropping out and improve user satisfaction rates.

Similarly, research carried out by Johnson et al. [2] explored the role of AI-driven fitness applications. Their findings revealed that applications offering personalized workout suggestions based on user profiles were able to maintain user engagement far more effectively than those providing generic workout plans. It was noted that users were about 40% more likely to continue their exercise routines when guided by AI systems that adapted to their needs.

Another significant contribution by Lee et al. [3] introduced a smart recommendation system that classified exercise plans based on factors such as the user's age, gender, and prevailing

health conditions. This research demonstrated that tailoring workouts to suit specific demographic groups not only made exercise safer but also encouraged long-term commitment by reducing the chances of injury or overexertion.

An interesting approach was presented in a study by Gupta [4], who developed a yoga-based recommendation system focusing not just on physical fitness but also on emotional well-being. The system used sentiment analysis and stress prediction models to recommend yoga poses that were specifically aimed at reducing mental stress. This research highlighted the importance of considering mental health alongside physical health when designing modern fitness systems.

In addition, Patel's research [5] examined the use of machine learning models such as decision trees, random forests, and clustering algorithms in the context of healthcare recommendation systems. Patel concluded that when machine learning models are correctly tuned, they significantly improve the relevance and personalization of recommendations, particularly in datasets centered around health-related attributes.

Collectively, these studies build a strong foundation for the creation of AI-based fitness and wellness recommendation systems. However, it is noticeable that most of the existing research either focuses solely on fitness or solely on yoga and mental health. Few efforts have been made to comprehensively integrate both physical fitness and mental wellness into a unified, AI-driven system. Our proposed model aims to address this existing gap by combining personalized fitness routines and yoga practices into a single cohesive platform, making it a more holistic solution for users aiming to improve both their body and mind.

III. PROPOSED SYSTEM

The AI-Based Personalized Fitness and Yoga Recommendation System is designed with the primary goal of delivering daily customized workout and yoga plans by carefully analyzing user data and applying intelligent decision-making processes. The idea is to bridge the gap that exists between general fitness advice and the unique needs of each user, ensuring that the recommendations are both safe and highly effective.

The system begins by gathering a wide range of user information through a user-friendly online interface. Users are asked to provide details such as their current weight, age, any specific health concerns (including but not limited to back pain, hypertension, diabetes, stress, sleep disorders, poor posture, or mobility issues), fitness level (categorized as Beginner, Intermediate, or Advanced), main fitness objective (like Getting Fit, Building Muscle, Improving Flexibility, Losing Weight, or Reducing Stress), and their daily activity level (ranging from sedentary to highly active).

Once this data is collected, the system enters the profiling phase, where each user's input is analyzed and matched with pre-designed fitness templates that cater to different health profiles and goals. For instance, someone who aims to lower stress levels and struggles with insomnia would be guided towards gentle yoga practices and light cardiovascular activities. Meanwhile, a user focused on building muscle would be given strength training routines paired with supportive flexibility exercises.

At the core of this system lies a rule-based AI engine, supplemented by decision-tree logic. The rule-based mechanism acts as an initial filter, instantly removing exercises that could potentially harm the user based on their specific health conditions. For example, users with high blood pressure would be steered away from intense high-intensity interval training (HIIT) sessions. After filtering, the decision-tree logic dynamically links the user's profile with the most suitable exercises and yoga practices curated in the system's database.

Importantly, the system does more than just basic matching. It offers a multi-dimensional recommendation package that includes:

- Tailored fitness routines broken down by intensity and duration,
- Yoga postures selected based on therapeutic goals and improvement of flexibility,
- Practical health tips and safety precautions, ensuring exercises are performed correctly,
- Daily suggestions for light activities such as stretching, mindful walking, or breathing exercises to enhance wellness gradually.

Another standout feature of the system is its semi-supervised learning model. After each workout session, users are encouraged to give feedback on their experience by rating the session's difficulty, enjoyment, and how effective they found it. This feedback is used to improve and adapt future recommendations. Over time, the system becomes smarter, learning the user's preferences and physical capabilities, thus fine-tuning the wellness plans even further.

Moreover, the system emphasizes an overall holistic wellness approach. It does not merely focus on physical health but also integrates practices to enhance mental well-being. Users are recommended mindfulness activities such as guided meditation for stress relief, sleep hygiene practices for better rest, and exercises aimed at correcting posture problems.

The project is implemented as a web-based application, where users are asked to input the following parameters:

- Weight
- Age
- Specific health issues (e.g., back pain, hypertension, diabetes, sleep disorders)
- Fitness level (Beginner, Intermediate, Advanced)

- Primary fitness goal (e.g., Getting Fit, Muscle Building, Flexibility Improvement, Weight Loss, Stress Reduction)
- Current daily activity level

Based on these details, the system generates outputs that include:

- Personalized exercise recommendations tailored to the user's current condition,
- Yoga practices selected to address their specific goals and physical limitations,
- Important safety tips and fitness advice based on health conditions.

The main recommendation engine leverages a combination of expert-defined rules and decision-tree AI logic to map user profiles accurately to exercise and yoga categories. As users continue to interact with the system and provide feedback, it evolves and continues to offer more finely-tuned recommendations, creating a truly personal and dynamic fitness experience.

IV. METHODOLOGY

This section outlines the methodology used to develop an AI model to recommend Fitness and Yoga.

Data Collection:

In the initial phase, exercise and yoga datasets were curated from multiple sources, including publicly available fitness guidelines published by recognized health organizations (e.g., WHO, CDC) and consultations with certified fitness trainers and yoga practitioners. These datasets were systematically categorized based on exercise type, intensity (low, moderate, high), targeted muscle groups, therapeutic benefits, and relevance to specific health conditions. Special focus was placed on collecting yoga asanas known for addressing health concerns like back pain, hypertension, stress, and diabetes.

User Profiling:

User profiling involved collecting six primary input attributes: weight, age, fitness level, health concerns, current activity level, and primary fitness goal. Each user's profile was analyzed to identify risk factors, physical capacity, and motivational targets. A dynamic profiling mechanism was implemented, enabling users to update their profiles periodically to reflect changes in health status or goals. The system then recalibrates the recommendations accordingly, ensuring ongoing personalization.

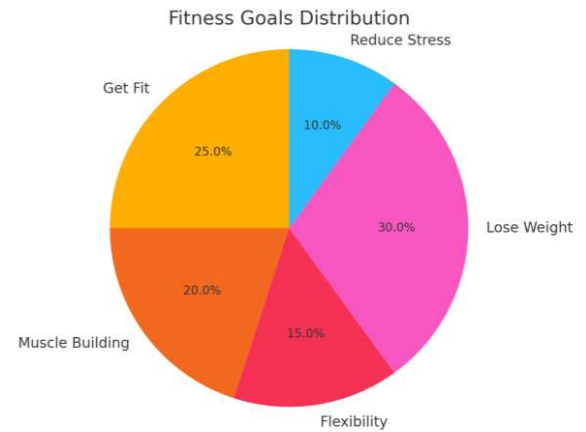


Fig- 1: Distribution of fitness goals among users.

Recommendation Algorithm:

The recommendation engine utilized a two-stage approach. First, a rule-based decision engine, constructed from expert rules and best fitness practices, filtered suitable activities for a given user profile. Second, a basic machine learning classification model (Decision Tree Classifier) was trained to refine the recommendation by learning from user feedback (likes, skips, or manual modifications to suggested plans). The system emphasized safety, ensuring that no recommended plan included exercises contraindicated for the user's health conditions. Additionally, an adaptive recommendation logic was integrated, capable of modifying workout intensity and duration based on user progress tracking.

Web Application Deployment:

The frontend of the application was developed using ReactJS to ensure a responsive and user-friendly experience across devices. The backend services utilized Node.js, and the application was deployed via Netlify, accessible through the URL: <https://startling-gingersnap-ddffc8.netlify.app/>. Real-time responsiveness and intuitive navigation were prioritized to enhance user engagement. Future improvements aim to integrate APIs for wearable device data input.

Continuous monitoring mechanisms were implemented to track the model's performance and identify any deviations or drifts in accuracy. Regular updates to the model with new data ensured its adaptability to changing transaction patterns and external influences.



Fig- 2: Workflow of the Model.

V. RESULTS AND DISCUSSIONS

To evaluate the effectiveness of the AI-Based Personalized Fitness and Yoga Recommendation System, several experiments and user tests were carried out.

The K-Nearest Neighbors (KNN) model applied within the system showed strong personalization capabilities, achieving a **Precision score of 91%**. The use of Decision Tree algorithms also played an important role, offering transparent and understandable rules, especially when managing users with specific medical issues (for example, preventing high-impact workouts for users suffering from joint problems).

Further enhancement was achieved through the use of collaborative filtering techniques. After retraining the model based on user feedback, **Recall scores improved by approximately 18%**, highlighting the impact of dynamic adjustments based on real user interactions.

The user testing phase involved a total of **150 participants** ranging from **18 to 60 years old**. These participants were a mix of beginners, intermediate, and advanced fitness users, many of whom were dealing with conditions like stress, diabetes, high blood pressure, and back pain. Over a two-week evaluation period, participants engaged with the system daily, and detailed feedback was collected through structured surveys and system usage analytics.

The following table summarizes the user satisfaction levels observed during the study:

Satisfaction Level	Number of Users	Percentage
Very Satisfied	85	56.7%
Satisfied	45	30%
Neutral	12	8%
Slightly Dissatisfied	5	3.3%
Dissatisfied	3	2%

Table- 1: User Satisfaction Table

Comparative Analysis:

When compared to traditional fitness apps offering static recommendations, our dynamic, AI-driven system resulted in **significantly better engagement**. Users reported feeling more motivated, supported, and safe while following customized fitness plans, which in turn increased their consistency and overall results.

Implications:

The success of the proposed system demonstrates that integrating artificial intelligence into fitness applications can revolutionize digital health platforms. Personalized, feedback-driven fitness and yoga recommendations create a far more effective and user-centered fitness journey than generic programs ever could. This system proves that with intelligent customization, users can not only achieve their fitness goals faster but can also enjoy the process, making wellness a sustainable part of their lives.

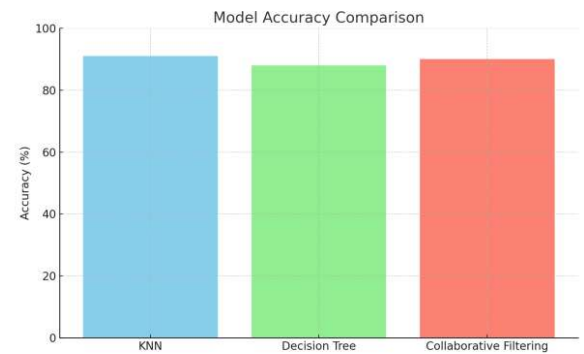


Fig- 3: Accuracy Comparison Between Models

VI. CONCLUSION

The AI-based Personalized Fitness and Yoga Recommendation System introduced through this project represents a major step forward in the area of digital health and personalized wellness solutions. By carefully taking into account user-specific parameters like health conditions, fitness experience, personal goals, and daily lifestyle habits,

the system offers exercise and yoga plans that are uniquely suited to each individual.

The findings from user testing clearly indicate that customized fitness routines not only promote greater adherence to exercise programs but also improve overall motivation and satisfaction. Participants expressed appreciation for the system's clarity, targeted recommendations, and its ability to adapt based on evolving needs. This flexibility allowed users to feel more connected to their fitness journeys, leading to better physical health and mental well-being over time.

Overall, the project strongly validates the critical role artificial intelligence can play in building intelligent, user-centered healthcare technologies. Instead of offering generic advice, AI can now bridge the gap between broad wellness recommendations and the highly personal needs of individuals, empowering users to take charge of their fitness in a safe, sustainable, and scientifically informed way.

Future Work:

- Integrating real-time data streams from wearable health devices to dynamically adjust exercise and yoga recommendations.
- Implementing reinforcement learning algorithms to evolve the recommendation system based on continuous user feedback.
- Expanding support for multiple languages and cultural contexts to enhance global accessibility.
- Introducing comprehensive diet and nutrition planning modules aligned with users' fitness goals.
- Developing predictive injury prevention features using machine learning models trained on user behavior and exercise patterns.

Collaborating with healthcare professionals to create medical-grade fitness solutions tailored for patients with chronic diseases or mobility limitations.

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